

HARRISON APPIAH, Ph.D.

Sustainable Materials Chemistry • Chemical & Environmental Engineering • Waste Valorization

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RESEARCH PROFILE

Chemist and materials/environmental researcher with a Ph.D. in Environmental Science (Material Chemistry) and an M.S. in Biomaterial Science (Chemistry). Research experience spans the chemical and materials valorization of municipal solid waste, solvent-targeted recovery of post-consumer plastics, polymer processing and biocomposites, biomass pyrolysis/biochar, and deep-eutectic-solvent conversion of lignocellulosic feedstocks to platform chemicals. Experienced in advanced chemical and materials characterization, laboratory method development, process optimization, and data-driven experimental research. Seeking a postdoctoral fellowship at the intersection of sustainable chemical manufacturing, circular materials, waste-derived feedstocks, and emerging conversion technologies.

RESEARCH INTERESTS

Sustainable chemical manufacturing • Chemical recycling & upcycling of waste plastics • Waste-derived chemical feedstocks • Biomass valorization & platform chemicals • Catalytic/electrified conversion pathways • Sustainable polymers & composites • Process optimization & characterization • Circular materials & resource recovery

RESEARCH EXPERIENCE

Graduate Research & Teaching Assistant | Renewable Materials Laboratory, University of Idaho, Moscow, ID | 2021–2026

- Conducted multidisciplinary research on municipal solid waste valorization, with emphasis on chemical recovery and materials applications of waste-derived plastics.
- Investigated solvent-based plastic extraction and recovery from heterogeneous municipal solid-waste streams and evaluated recovered polymers for downstream materials applications.
- Developed and evaluated waste-derived polymer/biocomposite systems using thermal, mechanical, rheological, spectroscopic, microscopic, and chromatographic methods to establish structure–property relationships.
- Conducted biomass-to-biochar research using pyrolysis/thermal conversion and evaluated carbon, nitrogen, fixed-carbon, yield, and material-property responses across processing conditions.
- Evaluated deep eutectic solvent systems for xylan-to-furfural conversion, integrating chemical conversion with thermal, spectroscopic, and rheological characterization.
- Designed experiments, analyzed experimental datasets, maintained detailed laboratory records, and communicated findings through research presentations and manuscripts.
- Supported undergraduate chemistry instruction and laboratory learning, including preparation of instructional materials and mentoring students in laboratory techniques and safe experimental practice.

SELECTED RESEARCH CAPABILITIES

Area	Techniques / Tools
Chemical & Analytical	GC, UHPLC/HPLC, ICP-MS, GC-MS/py-GC-MS, UV-Vis, FTIR, XPS
Materials Characterization	DSC, TGA, DMA, TMA, SEM, AFM, XRD, BET, density/pycnometry
Polymer / Materials Processing	Melt extrusion, injection molding, cast-film processing, rheology, polymer composites
Research Methods	Method development & validation, process optimization, experimental design, data analysis, quality control

ADDITIONAL PROFESSIONAL EXPERIENCE

Manufacturing Specialist | Schweitzer Engineering Laboratories (SEL), Pullman, WA | 2026–Present

- Assemble precision electronic and electromechanical components according to engineering drawings, work instructions, and standard operating procedures.
- Follow ESD controls, Lean Manufacturing, 5S, safety, and quality practices in a high-reliability electronics manufacturing environment.
- Collaborate with engineering, technical, and quality personnel to troubleshoot production issues and improve process efficiency.

Laboratory Technician (Chemistry) | St. Philip's College, San Antonio, TX | 2021–2022

- Prepared laboratory materials, cultures, and chemical solutions while maintaining laboratory standards and safe practices.
- Operated SEM, UV-Vis, pH meters, analytical balances, and related laboratory equipment; supported equipment troubleshooting and maintenance.
- Developed and validated analytical methods involving HPLC, GC, and SEM for chemical sample characterization.

Laboratory Research Chemist | Biomaterial & Analytical Chemistry, West Virginia University, Morgantown, WV | –May 2021

- Supported biomaterials and analytical chemistry research through experimentation, data management, documentation, and laboratory coordination.
- Maintained detailed research records and contributed experimental and technical support to project objectives.

EDUCATION

Ph.D., Environmental Science — Material Chemistry | University of Idaho | GPA: 3.78 | May 2026

M.S., Biomaterial Science (Chemistry) | West Virginia University | GPA: 3.66

B.S., Chemistry | Kwame Nkrumah University of Science and Technology | GPA: 3.1

Additional Training: Certificate in Semiconductor Fabrication — University of Texas at Austin / Intel Corporation

SELECTED PUBLICATIONS & RESEARCH OUTPUTS

1. **Appiah, H.**, Asamoah, P., & McDonald, A. (2026). Waste-to-energy technologies and their role in municipal solid waste management. *Recycling*, 11(3), 56.
2. **Appiah, H.**, Bar-Ziv, E., Klinger, J., & McDonald, A. (2026). Plastic recovery from municipal solid waste by solvent extraction. *ACS Sustainable Resource Management*, 3(2), 524–533.
3. **Appiah, H.**, Bar-Ziv, E., Klinger, J., & McDonald, A. (2025). Exploring new applications of municipal solid waste. *Sustainability*, 17(8), 3719.
4. Ewurum, N., Yusuf, S., **Appiah, H.**, & Tongco, J. V. (2025). Smart food packaging and solutions. In *Polymers and Composite Materials for Packaging* (pp. 273–296). Springer Nature.
5. **Appiah, H.**, Ewurum, N., Yusuf, S., & Tongco, J. (2025). Smart food packaging and solutions. In *Polymers and Composite Materials for Packaging* (pp. 395–413). Springer Nature.
6. Yusuf, S., Ewurum, N., **Appiah, H.**, & Tongco, J. (2025). Industrial hemp as precursor for sustainable bioproducts: Recent trends and prospects. *Fibers*, 13(11), 155.
7. **Appiah, H.**, McDonald, A. G., Oluwatosin, O., & Dawson-Andoh, B. (2025). Deep eutectic solvents for xylan-based furfural synthesis. *ACS Sustainable Resource Management*, 2(3), 536–545. <https://doi.org/10.1021/acssusresmg.4c00514>
8. Struhs, E., Mirkouei, A., **Appiah, H.**, & McDonald, A. (2024). Examination of in situ and ex situ catalytic fast pyrolysis and liquid fractionation utilizing a free-fall reactor. *Frontiers in Industrial Microbiology*, 2, 1426067.
9. **Appiah, H.** (2021). *Valorization of xylan in agroforestry waste streams* (Master's thesis). West Virginia University.
10. Sánchez-Rivera, K., Granger, C., **Appiah, H.**, Nelson, K., Grey, S., Sun, D., Estela-García, J., Chen, E., Xu, Z., & Osswald, T. (2024). Cast film production with polyethylene recycled from a post-industrial printed multilayer film by solvent-targeted recovery and precipitation. *ACS Materials Letters*, 6(9), 4042–4050.
11. Korang, J., Obiri, B., **Appiah, H.**, & Awuku, S. (2015). Calorific values and gravimetric yield of six wood fuel species in the forest transition zone of Ghana. *Ghana Journal of Forestry*, 31, 51–61.
12. **Appiah, H.**, Tongco, J. V., Yusuf, S. B., & Marquez, K. P. (2026). Engineering and development of high-entropy alloys. In S. Sandeep & V. Akarsh (Eds.), *Multi-scale mechanics of high entropy alloys* (pp. 1–23). Springer Nature.
13. **Appiah, H.**, Park, S. H., & Tongco, J. V. (2026). Advancements in green pretreatment, thermochemical conversion, and by-product valorization of lignocellulosic biomass for energy applications. *Journal of Carbon Research*.
14. **Appiah, H.**, Dawson-Andoh, B., Oginni, O., & McDonald, A. (2025). Evaluation of deep eutectic solvents for xylan-based furfural synthesis. *ACS Sustainable Resource Management*, 2(3), 536–545.
15. Huber, G., Ra, E., Cortes-Pena, Y., Rivera-Kohr, D., Callen, L., Zhou, P., Yu, J., ... **Appiah, H.**, Granger, C., Saha, N., & Klinger, J. (2026). Upcycling municipal solid waste to polymers and bioethanol.

CURRENT RESEARCH

My research develops next-generation approaches for converting heterogeneous waste and biomass feedstocks into value-added chemicals and functional materials. I am particularly interested in catalytic and electrified chemical-conversion pathways that enable selective, energy-efficient, and scalable processing of complex waste streams. My work builds on expertise in plastic waste valorization, polymer and biomass characterization, reaction and process optimization, and sustainable materials development, while expanding toward catalytic reaction engineering, electrified chemical manufacturing, and circular-economy technologies. The overarching goal is to bridge fundamental chemistry with scalable process development for sustainable chemical and materials manufacturing.